

TNA as a Structural Meta-Language: Mapping Quantum Information Panpsychism into the Theory of Axiomatic Necessity

Abstract

The Theory of Axiomatic Necessity (TNA) was originally developed as a metamathematical framework demonstrating the structural necessity of non-derivable axioms in every coherent formal system. From this theorem emerges a broader structural language capable of describing the relationship between description, realization, and ontological grounding.

This paper argues that TNA functions as a **meta-language** rather than a competing ontology. Instead of proposing what consciousness or reality ultimately are, TNA provides a structural grammar through which different ontological models can be expressed.

As a case study, we analyze the Quantum Information Panpsychism (QIP) proposed by Federico Faggin and Giacomo Mauro D'Ariano. We show that its central claims admit a natural translation into TNA without modifying either framework.

The resulting correspondence suggests that TNA is not an alternative theory of consciousness but a structural language capable of unifying apparently unrelated philosophical and scientific approaches.

1. Introduction

Theories of consciousness generally compete at the ontological level.

Materialism, panpsychism, idealism, dualism and information-based approaches disagree primarily about what reality fundamentally consists of.

TNA approaches the problem from a different direction.

Instead of asking **what exists**, it asks:

How must any coherent ontology be structurally organized?

This change of perspective transforms TNA from an ontological proposal into a structural grammar.

Its symbols describe relations rather than substances.

Consequently, multiple ontologies may be represented within the same formal language.

2. The Structural Language of TNA

TNA distinguishes three structural domains.

2.1 Unlimited Structural Possibility

$$N_{\infty}$$

The domain of unlimited structural coherence.

It represents the space of possible coherent structures independently of their finite realization.

2.2 Descriptive Closure

$$N_0$$

The domain of formal description.

Examples include

- mathematical theories,
- computer programs,
- neural dynamics,
- symbolic representations,
- physical equations.

Everything in N_0 is describable.

Nothing in the definition of N_0 guarantees realization.

2.3 Instantiated Reality

$$N_1$$

The domain of execution, embodiment and lived realization.

Here structures are not merely represented.

They exist as instantiated processes.

3. The Structural Corollary

The metamathematical theorem of TNA establishes the existence of structurally indispensable axioms for every coherent formal system.

Introducing the distinction between descriptive systems and instantiated systems immediately yields the following structural corollary.

Structural Non-Derivability Corollary

Let

- N_0 denote a complete axiomatic description of a system.
- N_1 denote the instantiated realization of that system.

Then

$$N_0 \not\vdash N_1$$

A complete description does not entail its own realization.

The transition from description to realization is therefore a structural boundary rather than an informational deficiency.

This corollary does not specify the ontology of realization.

It merely establishes that realization cannot be derived solely from descriptive closure.

4. Quantum Information Panpsychism

Federico Faggin proposes that consciousness is ontologically primary.

According to Quantum Information Panpsychism,

- the brain does not generate consciousness,

- consciousness precedes computation,
- physical systems operate as interfaces through which conscious entities interact with the observable world.

Within this framework,

the brain behaves as a quantum-classical computational machine,

while subjective experience belongs to a fundamentally different level of reality.

5. Structural Translation into TNA

The correspondence between both frameworks can be expressed directly.

QUANTUM INFORMATION PANPSYCHISM	TNA
Brain	N_0
Neural computation	N_0
Physical laws	N_0
Symbol manipulation	N_0
Qualia	N_1
Subjective realization	N_1
Conscious field	possible interpretation of N_∞

Under this translation,

Faggin's principal philosophical claim becomes

$$N_0 \not\vdash N_1$$

The computational structure of the brain cannot derive the existence of subjective realization.

Rather,

subjective realization constitutes an instantiated domain that exceeds descriptive closure.

6. Why TNA Is More General

This correspondence should not be interpreted as evidence that Quantum Information Panpsychism is implied by TNA.

The logical direction is the opposite.

TNA remains ontologically neutral.

It requires only the structural distinction

$$N_0 \neq N_1$$

together with the non-derivability relation

$$N_0 \not\vdash N_1.$$

How N_1 should ultimately be interpreted remains open.

Possible interpretations include

- phenomenology,
- embodied cognition,
- quantum consciousness,
- process ontology,
- idealism,
- panpsychism,
- or future theories not yet formulated.

The structural language remains unchanged.

Only the semantic interpretation varies.

7. TNA as a Meta-Language

The role of TNA is therefore analogous to the role played by category theory in mathematics.

Category theory does not determine the nature of mathematical objects.

Instead,

it provides a language describing relationships that remain invariant across different mathematical disciplines.

Similarly,

TNA does not determine the ontology of consciousness.

It specifies the structural grammar governing the relationship between

- possibility,
- description,
- necessity,
- realization,
- and exteriority.

Different ontologies become different semantic realizations of the same structural language.

8. Structural Invariance

Once expressed in TNA,

apparently unrelated philosophical theories reveal a common invariant.

Frank Jackson's Knowledge Argument

$$N_0 \not\equiv N_1$$

Quantum Information Panpsychism

$$N_0 \not\equiv N_1$$

Artificial consciousness

$$N_0 \not\equiv N_1$$

AI Alignment

$$N_0 \not\equiv N_1$$

Structural ontology

$$N_0 \not\equiv N_1$$

The theories disagree about the meaning of the symbols.

They preserve the same structural relation.

9. Conclusion

TNA should not be understood as one more theory of consciousness.

Rather,

it constitutes a structural meta-language capable of representing multiple ontological frameworks through a common formal grammar.

The metamathematical theorem establishes the necessity of structural exteriority.

The Structural Non-Derivability Corollary

$$N_0 \not\equiv N_1$$

extends this result to the relationship between description and realization.

Quantum Information Panpsychism provides one semantic interpretation of this grammar.

Other theories may provide different interpretations.

The formal structure remains invariant.

This invariance suggests that TNA occupies a higher level of abstraction than the ontological theories it represents.

Rather than competing with them,

it offers the structural language in which they can be compared.